

RailVision Strategic Positioning & Survival Brief – 2026

RailVision Strategy Brief

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CONFIDENTIAL STRATEGIC ANALYSIS

1. Executive Summary (BLUF)

RailVision is at a strategic crossroads. The core EcoRail / fuel-optimization capability is real, but the business is behaving like a feature in search of a platform, not a fully capitalized software company.

Bottom line: Based on current burn (\$200K+/month), \$0 revenue, and a 6–12 month runway subsidized ~40% by government funds, RailVision must treat 2026 as a survival and positioning year. The rational strategy is to operate as a product optimizing for strategic exit or deep strategic partnership, while executing a focused six month plan to (a) prove r shortline segment (6–10% target range on eligible trips) and (b) convert that proof into \$200–400K ARR and credible growth momentum.

Key decisions for the board in 2026:

- Accept that fuel optimization is a feature of a broader rail operations stack, not a standalone platform business at current scale and capitalization.
- Prioritize strategic sale / partner bundling options (with companies LB Foster, or ecosystem partners like First Analytics) over raising \$5–10M to build a full-stack operations platform.
- Commit to an execute or die six month survival plan (GWRR pilot, OmniTrax, and 1–2 additional proof points) with explicit go/no go t
- Use Jamie O'Rourke's and Dev Jain's strengths surgically: Jamie for access and positioning; Dev for technical credibility and pilot success—not as substitutes for missing commercial and implementation muscle.

If RailVision cannot reach \$100K+ MRR ("H\$1.2M ARR) or a clear stra December 2026, the board should plan for an acqui hire or controlled continued dilution with low probability of scale.

2. Market & Competitive Landscape

This section summarizes the relevant rail market structure, RailVision's target wedge, and the competitive benchmark set by Wabtec's Trip Optimizer and related systems.

2.1 Rail Market Overview

- North America locomotive base: ~29,500 units (~25,000 US, 3,000 Canada)
- Class I railroads: 7 carriers with large, modern fleets (e.g., UP ~7,300, BNSF ~6,000, NS ~4,000, CSX ~3,800, CN ~2,000 US units, CPKC ~1,600, KCS ~950) and annual fuel spend of \$9.4–11.8B.
- Shortline operators: ~15,000 locomotives, predominantly Tier 0 / 1, with limited capital and sparse digital tooling.
- Passenger & transit operators: ~1,200 locomotives (e.g., Metra, VIA, other regional systems) with mixed technology stacks.

RailVision's current strategic focus is the Shortline segment, where locomotive age, variability in crew skill, and lack of embedded energy management systems create a large market for a locomotive agnostic, tablet first solution.

2.2 Wabtec Trip Optimizer & Energy Management Stack

- Trip Optimizer (Wabtec):
- Installed base: ~12,000 locomotives globally; deployed across all North American Class I railroads.
- Capabilities: dynamic throttle and dynamic brake control, smart crew integration with multiple control systems (PTC/ATP, LOCOTROL, Movement Planner), Zero to Zero operation, network pacing, air brake advisement.
- Certified fuel savings: ~10% EPA certified, with a broader claimed savings on service and territory.
- Price anchor: inferred \$15K–\$30K per locomotive (from a ~\$74M Trip Optimizer + LOCOTROL upgrade order), typically bundled with hardware, track databases, and deployment services.
- Positioning: energy management and automation feature set within locomotive and digital intelligence platform, not a standalone company.
- SmartHPT & LOCOTROL (Wabtec):
- SmartHPT optimizes horsepower per ton; EPA certified 5% fuel savings; Movement Planner, and Network Control.

- LOCOTROL automates distributed power control and integrates tightly with Trip Optimizer.
- Transportation Management System (TMS):
- Manages movement of >9M annual carloads; used by 520 shortlines.
- Critical insight: Wabtec serves shortlines digitally via TMS, but Trip Optimizer is not marketed or evidenced as a shortline solution.

2.3 Competitive Implications for RailVision

- Feature vs platform: The market validates that fuel optimization and energy management are high value features inside larger stacks (Wabtec Digital Intelligence not self standing platforms).
- White space in shortlines:
- Trip Optimizer and SmartHPT are tuned and priced for Class I fleets and modern power, not for Tier 0 shortline fleets.
- Shortlines typically lack PTC/ATP, LOCOTROL, and capital to retrofit with \$15K–\$30K/locomotive systems.
- This creates space for tablet based, locomotive agnostic advisory economics in the \$300/month per locomotive range.
- Validation partners:
- LB Foster: Friction management programs (top of rail, wayside app) with *fuel economy and performance validation* outcomes.
- First Analytics: Designs measurement systems and statistically valid technologies (e.g., DiD, regression with covariates, interrupted time series, bootstrapping).
- Together they provide a playbook RailVision can emulate for proof of savings in shortlines.

Conclusion: the competitive benchmark confirms the technical legitimacy and economic value of fuel optimization, but also underscores that RailVision must either (a) attach itself to a broader platform or (b) become the shortline native feature that an acquirer adds to their stack.

3. RailVision Today: Position, Technology, Economics

This section summarizes RailVision's current strategic position using only provided facts and context.

3.1 Company Snapshot

- Legal name: RailVision Analytics Inc.
- Industry: Rail.
- Website: railvision.ca.
- Years in operation: ~8 years.
- Capital invested: \$11M.
- Role in deals: Principal vendor; owns the customer relationship and controls SOW, pricing, and renewals.
- Strategic thesis: "RailVision is taking a staged, revenue generating foundational platform of autonomy," starting with EcoRail (a "Google AI enabled, fuel optimization product").

3.2 Product & Technology Focus

- EcoRail / RailVision product:
- Positioned as a locomotive agnostic, AI enabled, in cab advisory system that uses self driving algorithms to recommend throttle and speed.
- Value proposition: fuel and GHG savings through better train handling "smart cruise control" for rail but delivered as a light touch overlay integration.
- Targeted savings: aligns with provided fuel savings metric of 5–10% on eligible trips, with expected range described as "high single digit" and values of 6%+.
- Focus segment: Shortline operators with predominantly Tier 0 locomotives and limited digital energy management capabilities.

3.3 Financial Trajectory & Cost Structure

- Revenue trajectory (historic):
- 2018: \$31K
- 2019: \$31K

- 2020: \$23K
 - 2021: \$173K
 - 2022: \$383K
 - 2023: \$768K
 - 2024: \$824K
 - 2025: \$0 (collapse to zero recognized revenue)
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- 2025 expense breakdown (planned/actual mix): Total "H" \$2.4M
 - R&D / Engineering: \$1.55M
 - Sales: \$162K
 - Marketing: \$10K
 - Product: \$6K
 - G&A: \$717K
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- Burn & runway context (supporting narrative):
 - Monthly burn: \$200K+.
 - Monthly revenue: \$0.
 - Government subsidies: cover ~40% of burn.
 - Implied runway: 6–12 months, depending on new capital and cost actions.

Implication: RailVision has credible IP and a working product, but the revenue engine stalled in 2025 while the company continued to invest heavily in engineering. The current cost base is sized for a more mature SaaS company than the revenue line supports.

3.4 Talent & Leadership

- Jamie O'Rourke – CEO (RailVision):
- Deep industry credibility and long standing relationships, including the GWRR executive team.
- Track record cited as having "overseen the prevention of more global derailments than anyone else in history" (from supporting context), making him an excellent spokesperson.
- Limitation: relationships are not directly convertible into scalable revenue without strong implementation, customer success, and sales infrastructure.

- Dev Jain – CEO and Founder (from pitch materials):
- Leads the technical vision for "building a fully autonomous rail industry" starting with a revenue generating, AI enabled solution.
- Has overseen \$4M+ in engineering spend (2023–2025) that produced real IP and a functioning system.
- Limitation: the hard problems now are deployment, integration, and change management in messy shortline environments, not core algorithm development.

3.5 Core Problem Statement

- The product works and targets a real, validated problem (fuel savings of 5–10%+ are economically meaningful; competitors have proven this at Class I scale).
- However, RailVision currently:
- Behaves like a feature in search of a platform, not a fully scaled company.
- Has insufficient commercial and implementation capacity to turn pilots into durable ARR.
- Faces a closing pivot window given burn, zero 2025 revenue, and the need to reach \$100K MRR by June 2026 to avoid a December 2026 hard landing.

This context anchors the strategic options that follow.

4. Strategic Options for RailVision

This section lays out the realistic strategic paths available to RailVision given its current product position, financial constraints, and market structure.

4.1 Option A – Build a Full Independent Platform Company

Description: Raise \$5–10M and invest 2–3 years to build out a comprehensive rail operations platform around EcoRail (planning, dispatching, TMS like functionality, integrations, etc.).

Requirements (from context):

- \$5–10M in new capital to fund:
- Commercial infrastructure: sales, marketing, customer success, implementation.
- Product expansion: beyond fuel optimization into full operations stack.
- 2–3 years of runway to reach \$20M+ ARR and justify a \$50–100M+ exit.

Pros:

- Full control of product and customer roadmap.
- Potential for larger standalone valuation if successful.

Cons (critical):

- Not viable under current burn and runway; would require a major, immediate change in position of weakness (2025 revenue collapse, \$0 current revenue).
- Direct competition with entrenched platforms (Wabtec Digital Intelligence, Class I TMS and movement planners) with far deeper balance sheets and installed bases.
- High execution risk in building a broad operations platform from a single feature starting point.

Assessment: This path is theoretically attractive but practically infeasible given RailVision's 2026 starting point.

4.2 Option B – Optimize as a Product for Strategic Sale

Description: Treat RailVision explicitly as a product looking for a company, and optimize the next 12–18 months for acquisition by a strategic player building a broader stack (e.g., Wabtec, Progress Rail, large integrators, or a TMS vendor) or by an aggregator (GWRR or OmniTrax) who wants embedded energy management.

Key moves:

- Prove that EcoRail reliably delivers 6%+ fuel savings in shortline environments, validated with rigorous analytics (DiD, regression, ITS) akin to First Analytics' playbook.
- Convert 2–4 lighthouse customers (e.g., GWRR pilot expansion, Vermont Rail Systems, OmniTrax subset) into \$1–2M ARR over 12–18 months.
- Document implementation playbooks, change management methods, so an acquirer can see low integration friction.

Pros:

- Requires \$2–3M bridge rather than \$5–10M growth round (per context).
- Plays to RailVision's strengths as a narrow but deep feature with credible IP.
- Aligns with how the market already consumes fuel optimization (as a feature inside larger stacks).

Cons:

- Inherently caps upside vs. building a standalone company.
- Requires discipline in focusing roadmap and go to market around w repeatability, attach rate), not long term vision.

Assessment: Based on the provided narrative, this is the recommended primary path.

4.3 Option C – Strategic Partner Bundling (Hybrid Path)

Description: Form deep, non exclusive partnerships where RailVision hardware from established players (e.g., LB Foster for friction management, First Analytics for validation, a TMS vendor for dispatch), effectively becoming the AI f a broader value proposition.

Examples:

- LB Foster + RailVision + First Analytics: A combined offering where LB Foster deploys physical friction management, RailVision provides operational advisory/fuel Analytics validates the combined savings.
- Shortline TMS provider + RailVision: Integrate RailVision's in cab (carloads, routes, schedules) to deliver holistic efficiency packages to the 520 shortlines already on Wabtec TMS or competing systems.

Pros:

- Faster distribution into shortlines through existing channels.
- Shared proof and validation responsibilities (easier for customers to trust outcomes).
- Enhances attractiveness for later strategic acquisition.

Cons:

- Requires careful contracting to ensure RailVision retains data rights and customer relationship influence.
- Risk of being commoditized or sidelined if partner priorities shift.

Assessment: This is a strong complementary strategy to Option B, not a replacement. Partner bundling can accelerate adoption and strengthen the case for an eventual strategic sale.

4.4 Option D – Status Quo / Incrementalism

Description: Continue as is: small pilots, heavy engineering spend, and no explicit orientation toward sale or platform build.

Assessment: The context is explicit—this is not a viable option. With \$200K+ monthly burn, \$0 revenue, and finite subsidies, drift leads to shutdown or distressed

Board implication: The board should explicitly endorse Option B (optimize for strategic sale) with Option C (partner bundling) as the distribution and validation accelerant, and decisively reject Option D. Option A should remain a theoretical path only if conditions dramatically change (e.g., unexpected large capital infusion plus early ARR traction).

5. Six-Month Survival Plan (Execute-or-Die)

This section condenses the detailed six month plan into an executive summary of key outcomes and decision gates.

Timeframe assumption: Starting January 2026.

5.1 Month 1 – Triage & Reset (January 2026)

Objectives: Understand what broke in 2025, right size the organization to meet the target.

Key actions:

- Produce an honest, written post mortem of the 2025 revenue collapse.
- Meet with the GWRR executive team to clarify blockers and whether a path forward exists.
- Quantify CAC, LTV, and churn using 2023–2024 data.
- Assess cash, burn, subsidies, and runway.
- Rebalance the team:
- Reduce engineering headcount by ~30%; redeploy capacity to implementation and customer success.
- Hire Implementation Lead (senior engineer who prefers field/customer work).
- Hire Sales specialist with shortline experience.
- Cut all discretionary spend (conferences, consultants, experimental marketing).

Deliverable:

- Revised 2026 plan with a realistic \$750K ARR target and quarterly milestones.

5.2 Month 2 – GWRR Do-or-Die (February 2026)

Objective: Secure a binding, high stakes pilot with GWRR or explicit commitment.

Key actions:

- Executive level meeting (Jamie + key leadership) with GWRR decision makers.
- Present a 3 property, 90 day pilot proposal with clearly defined success metrics.
- Validated 6%+ fuel savings vs. recalibrated baseline on eligible trips.

- High crew acceptance and low opt out rates.
- Agreement to expand to 10+ properties on success.
- Offer economics that make the decision easy: 25% of verified savings (not 30%) or equivalent value based pricing.
- Select properties with highest probability of success (avoid the hardest use cases initially).
- Assign a dedicated internal tiger team for the pilot.

Deliverable:

- Signed GWRR pilot agreement *or* explicit board acknowledgment that GWRR is off the table, triggering Plan B (focus on other aggregators and shortlines).

5.3 Month 3 – Implementation Discipline (March 2026)

Objective: Execute flawlessly on the first GWRR property and prepare

Key actions:

- Conduct site visits and assessments for all 3 GWRR pilot properties.
- Provision and validate hardware and connectivity (tablets, brackets, data links).
- Lead engineer training, ideally with Dev Jain present for credibility.
- Deploy EcoRail on Property #1, with:
- Daily monitoring by Implementation Lead.
- Clear issue tracking and rapid response.
- Weekly operational check ins with GWRR.
- In parallel:
- Prepare Vermont Rail Systems deployment for April.
- Begin OmniTrax discovery and define a realistic 3–5 property pilot concept (not 19+).

Deliverable:

- Property #1 live, stable, and generating early fuel savings signals

5.4 Month 4 – Validation & Expansion (April 2026)

Objective: Prove that fuel savings are real, statistically valid, and expandable.

Key actions:

- Deploy EcoRail on Properties #2 and #3 (GWRR).
- Produce a 30 day fuel savings report for Property #1, using robust
- Difference in differences vs. control locomotives/routes where pos
- Regression controlling for consist, tonnage, grade, speed profile, weather.
- Interrupted time series if controls are weak.
- Target: "e 6% documented fuel savings with confidence intervals.
- Have GWRR finance independently validate the results.
- Present results to GWRR executives and negotiate expansion to 10+ additional properties, per original design.
- In parallel:
- Deploy Vermont Rail Systems.
- Advance OmniTrax to a specific pilot proposal.

Deliverable:

- Signed GWRR expansion agreement with a credible path to \$200–300K ARR by June 2026.

5.5 Month 5 – Sales Engine Activation (May 2026)

Objective: Turn GWRR proof into broader market traction.

Key actions:

- Launch targeted outbound to aggregators and influential shortlines:
- Core narrative: "See how GWRR is saving 6%+ on fuel costs with EcoRail in legacy fleets."
- Arrange site visits and reference calls with GWRR for top 5 prospects.
- Finalize OmniTrax pilot agreement (3–5 properties, staged rollout).
- Close Vermont Rail Systems as first full production customer (beyond pilot) if results meet thresholds.
- Begin expansion deployment at GWRR properties 4–13.

Deliverable:

- \$300–400K ARR in signed deals and 3–5 customers in active deployment/pilot.

5.6 Month 6 – Funding or Exit Decision (June 2026)

Objective: Use momentum to secure either bridge funding or a strategic transaction path.

Key actions:

- Prepare investment materials: updated deck, financial model, customer proof, case studies (GWRR, Vermont Rail, early OmniTrax data).
- Run two tracks in parallel:
- Fundraising track: target \$2–3 M bridge from strategic or sector focused investors.
- Acquisition track: test appetite among strategic acquirers (e.g., Wabtec, Progress Rail, Alstom) and aggregators (GWRR, others).
- Align with board on decision framework:
- "e \$400K ARR and strong momentum: prioritize bridge financing, aim for strategic sale in 12–18 months.
- \$200–300K ARR with mixed execution: consider earlier strategic sale in the \$15–20M range.
- < \$200K ARR or weak proof: prepare for acqui hire or controlled sale.

Deliverable:

- Either capital secured to extend runway or a term sheet / active process for strategic transaction.

Board use: This six month plan is the minimum bar for treating 2026 year. Deviations should be explicit board decisions, not drift.

6. Recommended Path, Metrics & Operating Guardrails

This section crystallizes the recommended strategy and the metrics that should govern decisions.

6.1 Recommended Strategic Path

Given the facts:

- Fuel optimization is a validated feature class product, not a stand alone scale.
- Wabtec and others already embed these capabilities in larger stacks.
- RailVision has limited runway, strong IP, and early but fragile commercial proof.

Recommendation:

- Operate explicitly as a product optimizing for strategic exit, not as a platform company.
- Execute the six month survival plan as the core operating plan for the next 6 months.
- Use partner bundling (Option C) to amplify reach and validation (LB Foster, First Analytics, TMS providers, shortline aggregators).

6.2 Key Metrics & Decision Thresholds

The following metrics should be reviewed monthly and at each board meeting:

1. Revenue & Pipeline

- Monthly Recurring Revenue (MRR):
- Target: \$100K MRR by June 2026 ("H\$1.2M ARR run rate).
- Intermediate: \$300–400K ARR signed by end of Month 5 (May 2026).
- Number of active paying customers:
- Target: 3–5 customers in production or late stage pilot by June 2026 (BNSF, Union Pacific, Rail, OmniTrax properties).

2. Product Performance (Fuel Savings)

- Verified fuel savings percentage on eligible trips:
- Target: 6%+ documented savings, within the 5–10% expected range.
- Method: statistically robust validation (DiD, regression, ITS) following industry best practices.

- Coverage:
- Number of locomotives and train miles under EcoRail advisement.

3. Adoption & Change Management

- Engineer/crew adoption rate on equipped trains:
- Target: "e70% active use, opt out"d10% once stabilized.
- Time to first train on new properties:
- Target: "d14 days from contract signature to first live revenue train

4. Cost & Runway

- Net monthly burn (after subsidies): trend downwards or flat while revenue grows.
- Runway months at current burn and booked revenue; board to act if projected runway < 6 months without credible financing path.

6.3 Operating Guardrails

To prevent strategic drift:

- No large engineering expansions without direct line of sight to revenue.
- Every major decision must answer: "Does this extend runway or generate revenue in the next 90 days?" If not, it should be deferred or cut.
- Partnership discipline:
- Maintain RailVision control over SOW, pricing, and renewals where possible.
- Secure rights to trip, usage, and baseline data needed to compute fuel savings and support an acquirer's due diligence.
- Avoid white label or deep OEM branding that erases RailVision's identity.

6.4 Success Definition by December 2026

RailVision should define success in 2026 as achieving at least one of the following outcomes:

1. Momentum Company (Bridge + Growth Path)

- ARR "e \$1.2 – 2.0M.
- 5–10 paying customers with documented savings.
- Clear strategic interest from multiple acquirers but choosing to delay sale for valuation.

2. Strategic Sale

- ARR "e \$200–400K, with strong proof of product value and attach p
- Completed or near completed strategic transaction in the \$15–40M treatment for team and IP.

3. Orderly Wind Down / Acqui Hire (if metrics not met)

- If neither of the above is in reach and runway is < 6 months, prioritize employee outcomes and IP preservation via acqui hire discussions.

Board ask: Explicitly endorse this metric framework and require that any deviation triggers a board level discussion, not silent drift.

7. Risks & Board Discussion Questions

This section surfaces the major risks to the recommended strategy and proposes specific questions for board and leadership discussion.

7.1 Key Risks

1. Execution Risk on GWRR and Early Pilots

- If the GWRR pilot is delayed, under resourced, or shows ambiguous results, it may be difficult to create the flagship case study needed to unlock other deals.
- Mitigation: over invest leadership attention and top talent in the first pilot. Pilot scope tight and success metrics clear.

2. Statistical and Data Quality Risk

- Fuel savings are inherently noisy; poor data quality (fuel allocation, GTM calculations, missing telemetry) can undermine claims.
- Mitigation: adopt First Analytics style methods (clear KPIs, baseline, controlled experiments, regression, ITS, bootstrapping) and document methods up front. Where possible, involve third party validation.

3. Commercial Capacity Risk

- The company historically over indexed on engineering vs. sales/CS. Implementation Lead and sales hire, scaling beyond a few properties could strain the team.
- Mitigation: sequence growth deliberately; cap concurrent pilots; coordinate deployments (e.g., LB Foster field teams) to extend reach.

4. Partner Concentration Risk

- Over reliance on a single aggregator (e.g., GWRR) or a single strategic partner that relationship stalls.
- Mitigation: maintain a small portfolio of anchors (GWRR, Vermont Rail, OmniTrax, at least one transit/passenger operator) and at least two independent channels.

5. Funding & Timing Risk

- The plan assumes that demonstrable ARR momentum by mid 2026 will drive investment or strategic interest. Macro conditions or investor appetite may be unfavorable.

- Mitigation: start light strategic dialogues early (Q1–Q2 2026) so that acquisition is not a last minute fire drill if financing falls through.

6. Strategic Drift / Vision Misalignment

- Founders and early team may emotionally resist framing RailVision as a product headed for strategic sale rather than a standalone autonomy company.
- Mitigation: align on narrative and incentives (e.g., transaction linked to autonomy roadmap as part of a larger platform post sale).

7.2 Board & Leadership Discussion Questions

1. Company vs. Product

- Do we explicitly agree that, based on current facts, RailVision is a product looking for a company, not a fully independent platform business?
- If not, what new evidence or capital commitments would change that assessment?

2. Survival Plan Commitment

- Are we prepared to commit fully to the six month execute or die plan with changes and spend cuts in Month 1?
- What would cause us to intentionally deviate from this plan, and how will such deviations be governed?

3. GWRR Dependence & Plan B

- What is our explicit Plan B if GWRR declines to proceed or stalls (by end of Month 2)?
- Which 2–3 alternative customers or aggregators become the new anchors in that scenario?

4. Partnership Boundaries

- For partner bundling (LB Foster, First Analytics, TMS providers, OEMs), what are our non negotiables regarding:
- Data ownership and access?
- SOW and pricing control?
- Branding and customer relationship rights?

5. Funding vs. Sale Timing

- Under what conditions would we prefer a \$2–3M bridge vs. an earlier \$15–20M strategic sale?

- What are the board's expectations for founder and investor outcomes under each scenario?

6. Talent & Leadership Structure

- Do we need a COO/President with hands on scaling and implementation to complement Jamie's relationships and Dev's technical leadership?
- If yes, how will we fund and prioritize that hire given the runway constraints?

Intended use: This section is designed as a board meeting agenda: each risk and question can be a discrete topic, with clear pre reads (this brief plus updated financials) and decisions recorded explicitly rather than implicitly assumed.